CSE3016 - AWS SOLUTION ARCHITECT

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MODULE III

Networking & Content Delivery – Amazon API Gateway, Amazon CloudFront, Direct Connect, AWS Networking and Content Delivery, Amazon VPC (Virtual Private Cloud), VPC Peering.

Amazon API Gateway

What is Amazon API Gateway?

- Amazon API Gateway is a fully managed AWS service used to create, publish, monitor, and secure APIs at any scale.
- It serves as the gateway between client applications and backend services, routing requests efficiently.
- It simplifies API management by reducing infrastructure overhead and automatically scaling with demand.
- Supports RESTful APIs, WebSocket APIs, and serverless applications with flexible and cost-effective options.

Key Features of Amazon API Gateway

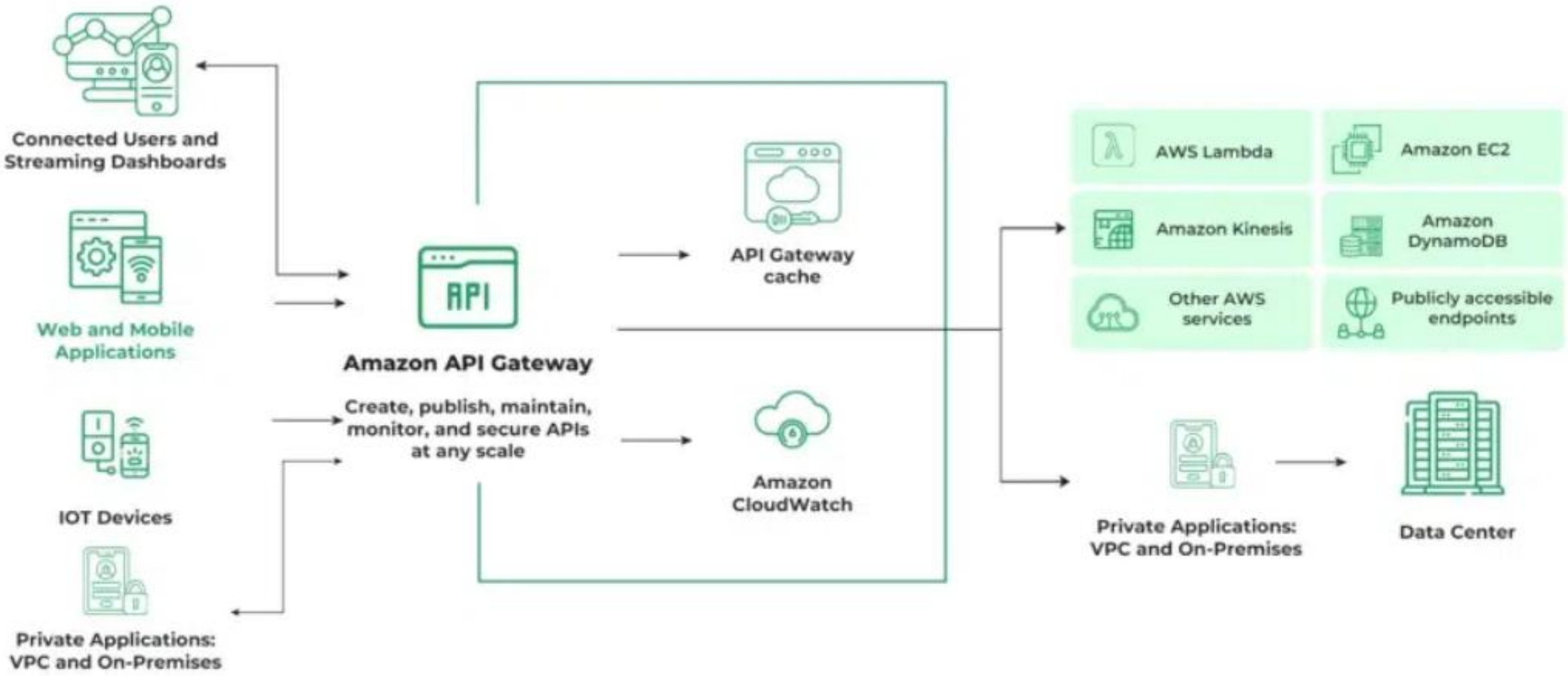
- **Traffic Management**
  - Provides load balancing, request throttling, and caching for optimal API performance under heavy traffic.
- **Security Features**
  - Offers authentication and authorization through IAM roles, custom authorizers, and Amazon Cognito integration.
- **Version Management**
  - Enables management of multiple API versions so different clients can access the correct version seamlessly.
- **Scalability**
  - Automatically scales to handle high-volume or sudden increases in API requests without downtime.
- **Monitoring and Analytics**
  - Integrated with Amazon CloudWatch to track performance metrics, request logs, and diagnose issues effectively.

How Amazon API Gateway Works?

- Amazon API Gateway acts as an interface between client applications and backend services. When a client sends a request, the API Gateway routes it to the appropriate backend service based on the API configuration. Once the backend processes the request, the service sends a response back to the API Gateway, which then passes it to the client.
- **Simplified flow:**

Client Request → API Gateway → Backend Service (AWS Lambda, EC2, etc.) → Response → API Gateway → Client

This architecture minimizes client-side complexity while centralizing API management.



Types of APIs Supported by Amazon API Gateway

- AWS API Gateway supports multiple API types to suit different application needs.
- **REST APIs (Public and Private)**
  - Follow the REST architectural style for building web services.
  - Provide a simple, flexible way to access data without heavy processing.
- **SOAP APIs**
  - Use the SOAP protocol for structured data exchange over XML.
  - Work across different platforms and operating systems using protocols like HTTP and SMTP.
- **GraphQL APIs**
  - Use GraphQL, an open-source query language and runtime for APIs.
  - Allow clients to request exactly the data they need, improving performance and flexibility.
- **WebSocket APIs**
  - Enable real-time, two-way communication between clients and servers.
  - Useful for applications like chat systems, live dashboards, and notifications.

Amazon API Management Components

- Following are some Amazon API Management Components that you need to be familiar with to get a better understanding.
- **HTTP (HyperText Transfer Protocol) API** is an application layer protocol that helps to communicate over the World Wide Web to get the data.
- **REST (Representational State Transfer) API** takes the HTTP standards to perform operations of **GET**, **POST**, **PUT**, **PATCH**, and **DELETE** on the data.
- **WebSocket** is a device communication protocol that provides point-to-point system communication channels over a single TCP. It enables stateful, full-duplex communication between client and server.

Benefits Of Amazon API Gateway

- **Simple to Use:** The AWS console provides very simple UI to use API GW which make its easy to manage and monitoring APIs.
- **Scalable:** Usually Amazon web services maintains very high scalable API gate ways then only they can handle the heavy demanding workloads.
- **High Secure:** API Gateway offers several capabilities, like custom authorizers and IAM integration, to help secure APIs.
- **Exposes Backend Services:** We can expose backend services to external clients, such web applications, mobile applications, and other APIs, with the use of API gateways.

Features of Amazon API Gateway

- **API creation and deployment:** Creating different types of APIs and managing them will be very easy.
- **Authorization and access control:** With the help of IAM roles, IAM policies, and custom authorizers you can achieve authorization and control mechanisms that API Gateway offers for your APIs.
- **Traffic management:** Load balancing, throttling and chaching are the some of the services offered by the API Gateway to managethe traffic.
- **API versioning:** Based on the request you API Gateway will route the traffic to the and to route traffic to the appropriate version based on the client's request.

Pricing of Amazon API Gateways (GW)

- You pay for use only, i.e you pay for the API calls you receive and the amount of data transfer. There is an optional data caching charged at an hourly rate that varies based on the cache size you select.
- Amazon also provides free tier services for **up to 12 months**, which includes:
- 1 million HTTP API calls
- 1 million REST API calls
- 1 million messages
- 750,000 connection minutes per month

Standard pricing of Amazon API Gateway:

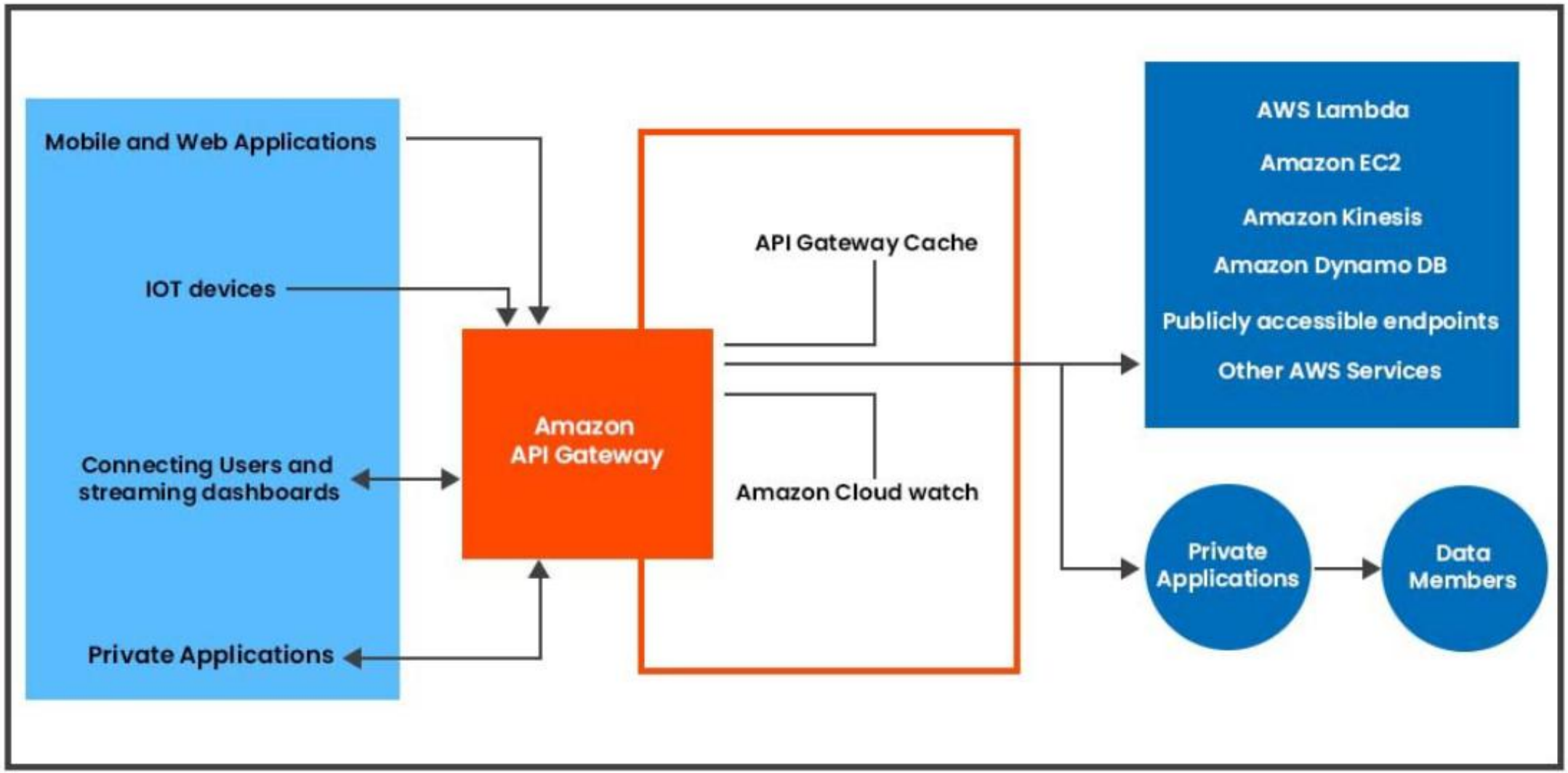
Free Tier:

- 1 million HTTP API calls
- 1 million REST API calls
- 1 million messages
- 750,000 connection minutes per month for 12 months

Pricing for the API Calls		
APIs	**No. of requests per month**	**Price(per million)**
HTTP	First 300 million	$1.00
	300+ million	$0.90
REST	First 333 million	$3.50
	Next 667 million	$2.80
	Next 19 billion	$2.38
	Over 20 billion	$1.51
WebSocket	First 1 billion	$1.00
	Over 1 billion	$0.80

Amazon API Gateway Architecture

- API Gateway provides a unified and consistent developer experience for building serverless applications on AWS.
- It acts as a gateway that connects client applications to backend services.
- Backend services can include Amazon EC2, AWS Lambda functions, or any custom web application.
- The architecture ensures secure, scalable, and efficient communication between clients and backend systems.


Mobile and Web Applications
IOT devices
Connecting Users and streaming dashboards
Private Applications
Amazon API Gateway
API Gateway Cache
Amazon Cloud watch
AWS Lambda
Amazon EC2
Amazon Kinesis
Amazon Dynamo DB
Publicly accessible endpoints
Other AWS Services
Private Applications
Data Members

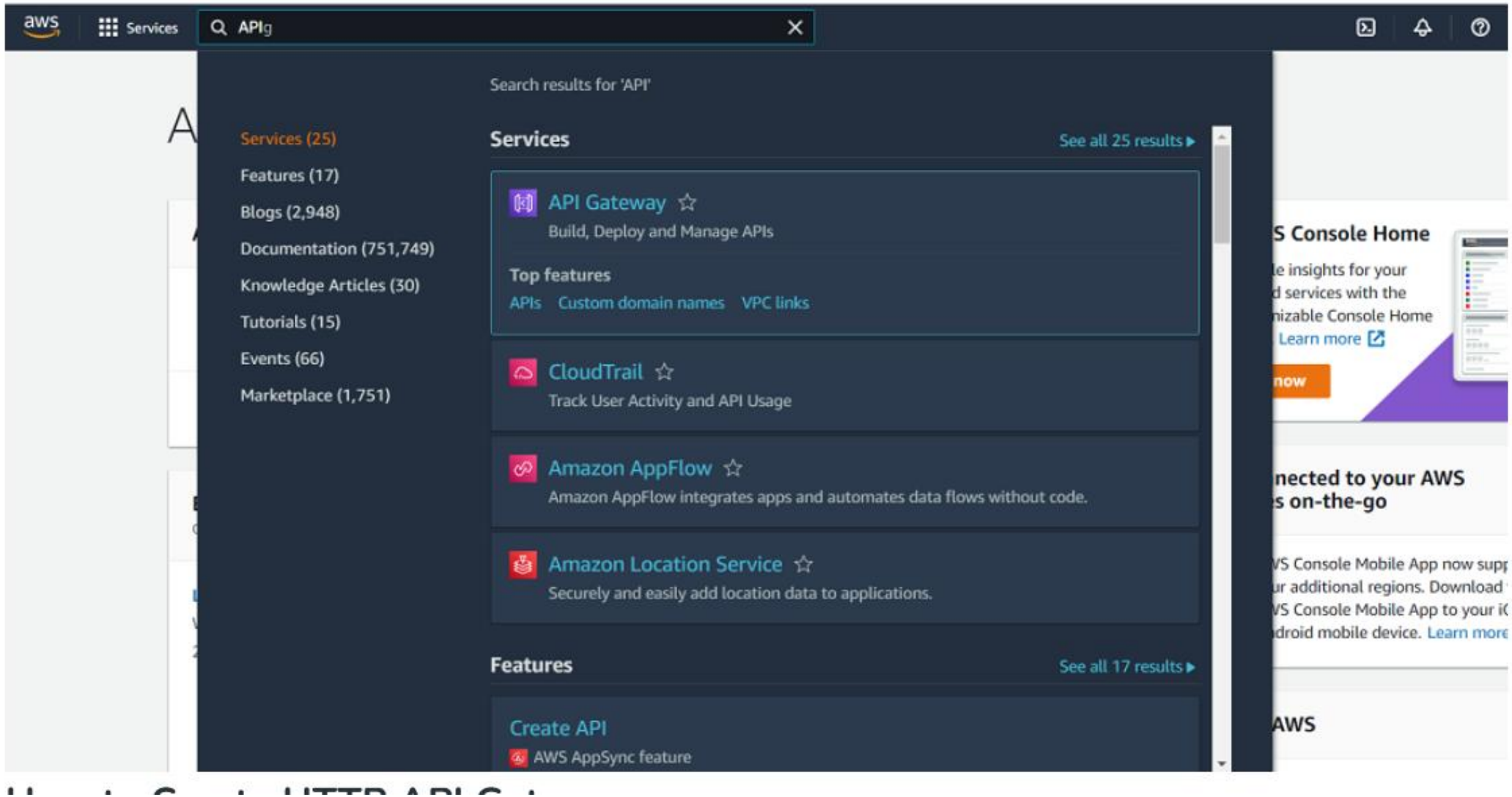
The entire architecture of AWS API Gateway consists of the below key components:

- **Amazon API Gateway:** It is used to create, publish, secure, maintain and monitor APIs.
- **API Gateway Cache:** Users can enable API caching to cache their endpoint responses, which can reduce the number of calls made to the endpoint and also improve the latency of API requests.
- **Amazon Cloud Watch:** It is a monitoring and observability service. It collects monitoring and operational data and visualizes it using automated dashboards, which allows users to visually monitor calls to their services.

- **Working With Amazon API Gateway**
- You can access **Amazon API Gateway** through the following tools:
- **AWS Management Console**
- **AWS SDKs**, including API Gateway V1 and V2 APIs
- **AWS Command Line Interface (CLI)**
- **AWS Tools for Windows PowerShell**

Example

- To create an HTTP API, you must first create an AWS Lambda function that will serve as the backend.
- After creating and configuring the Lambda function, you can set up an HTTP API using Amazon API Gateway.
- Once the API is created, you can test it directly from the console.
- **Steps to Get Started in the AWS Management Console:**
- Log in to your AWS account and open the AWS Management Console.
- Click on **Services** and search for **API Gateway**.

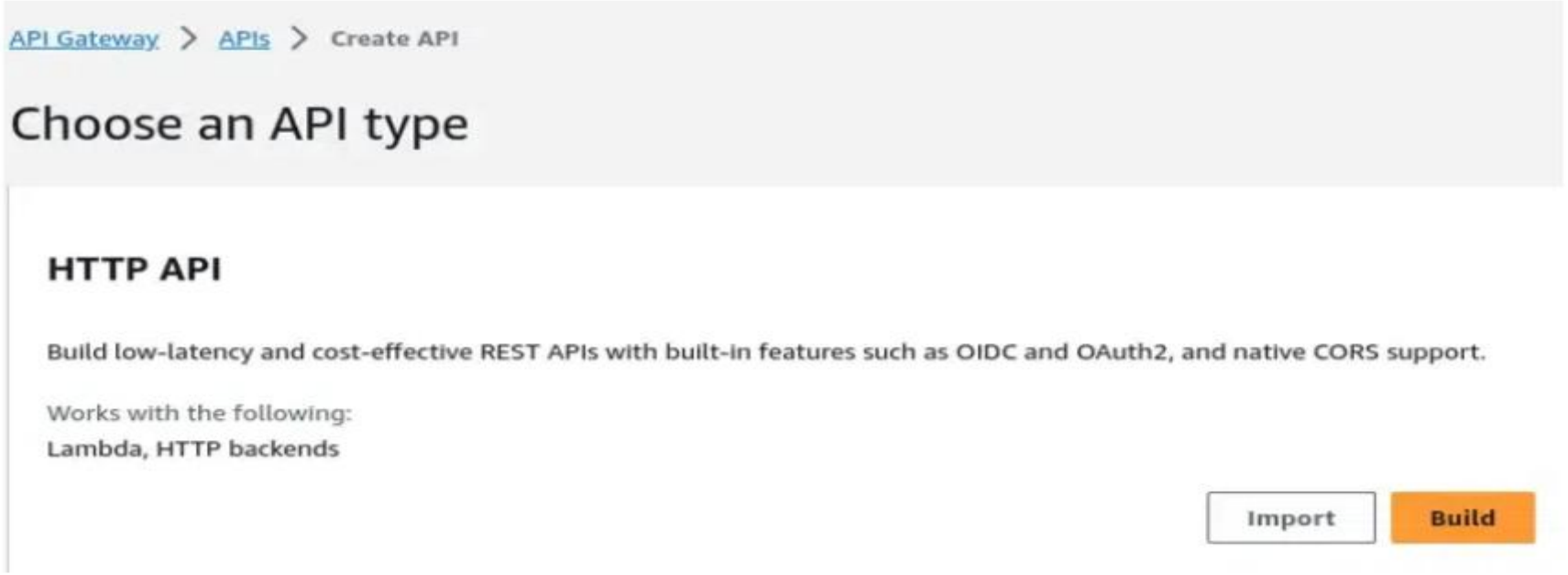

aws
Services
APIg
Search results for 'API'
Services (25)
Features (17)
Blogs (2,948)
Documentation (751,749)
Knowledge Articles (30)
Tutorials (15)
Events (66)
Marketplace (1,751)
Services
See all 25 results
API Gateway
Build, Deploy and Manage APIs
Top features
APIs
Custom domain names
VPC links
CloudTrail
Track User Activity and API Usage
Amazon AppFlow
Amazon AppFlow integrates apps and automates data flows without code.
Amazon Location Service
Securely and easily add location data to applications.
Features
See all 17 results
Create API
AWS AppSync feature
S Console Home
Learn more
nected to your AWS
s on-the-go
AWS

How to Create HTTP API Gateway

- Here’s a step-by-step guide to creating an HTTP API Gateway:
- HTTP API is an protocol which is used to communicate between the cline and servers.

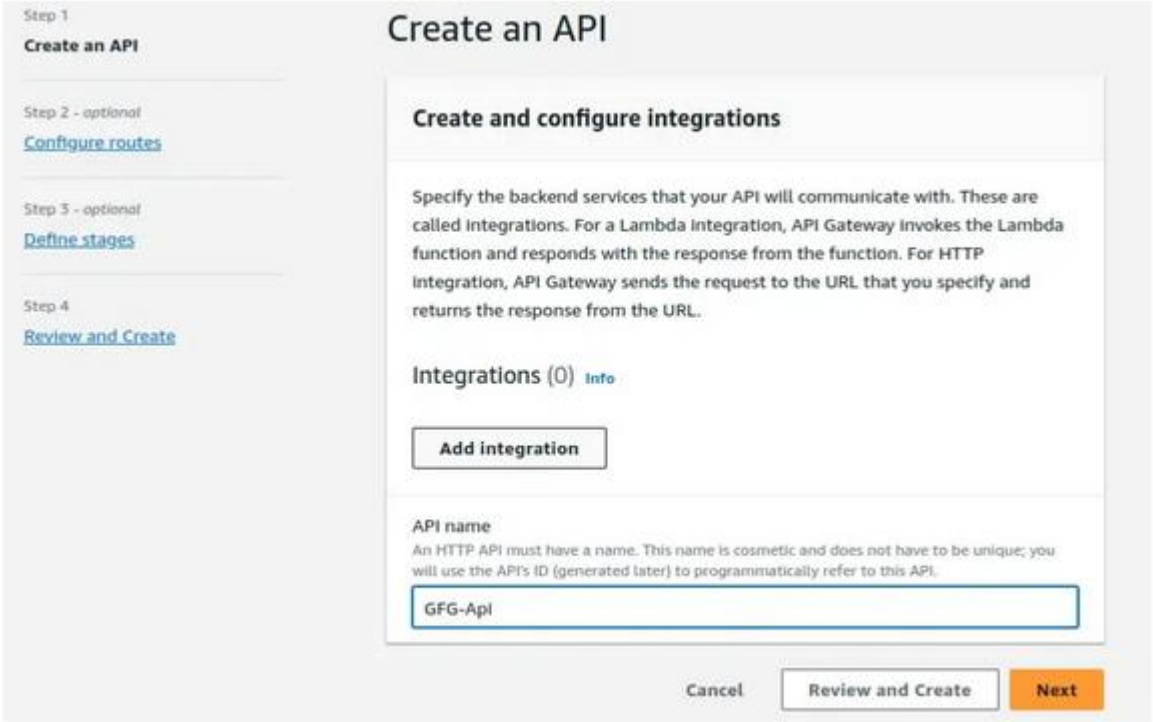
Step 1: Choose API Type

- Select the type of API that best fits your organization’s requirements. For this demonstration, we will choose **HTTP API**.



Step 2: Select Integration

- Choose the backend service you want to integrate with the API. For instance, API Gateway can invoke the Lambda function you've created earlier and return the function's response to the client.

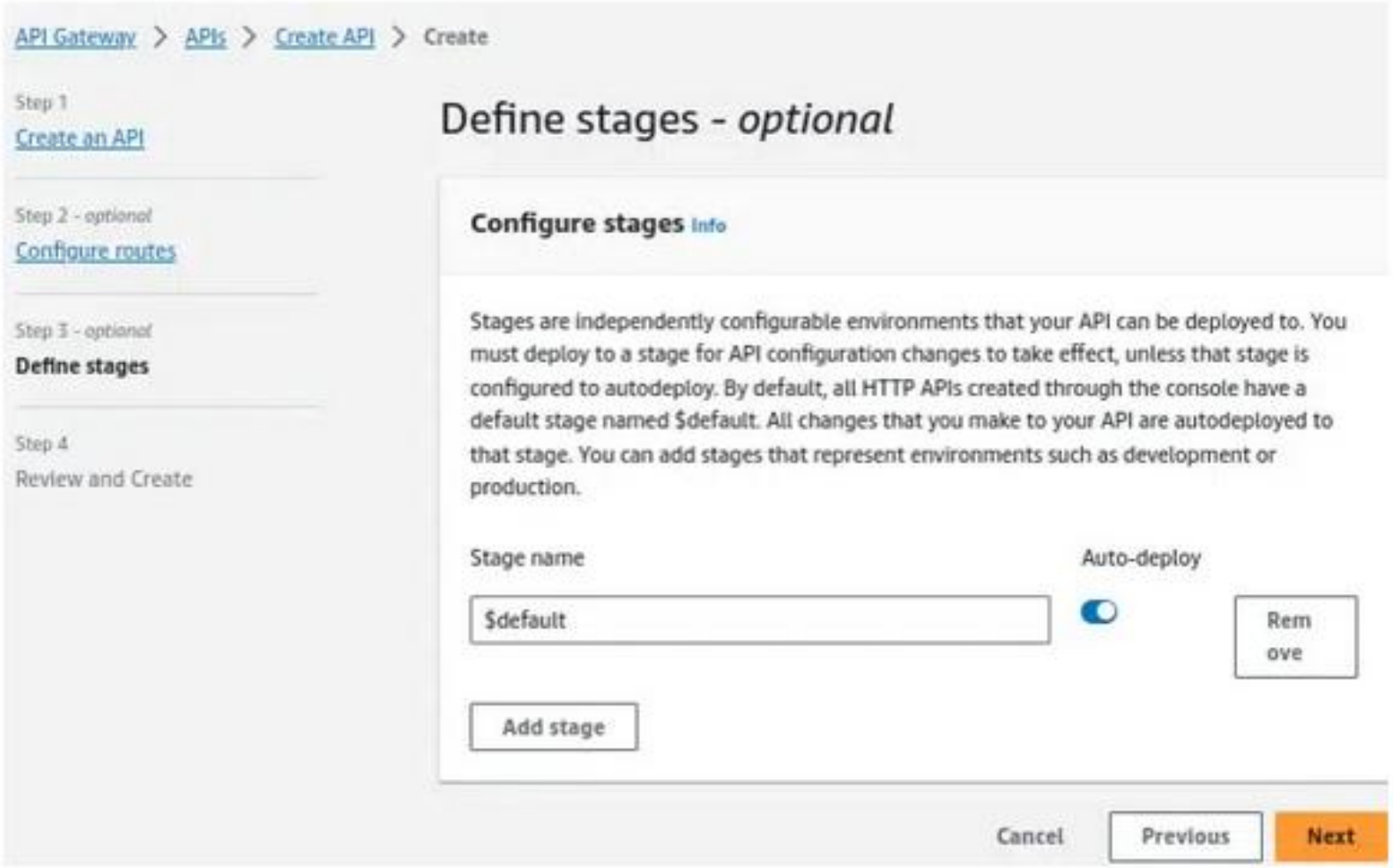


Step 3: Define Routes and Methods

- API Gateway uses routes to expose integrations to end users. Choose the HTTP method (GET, POST, PATCH, HEAD, etc.) based on the needs of your organization and how the API will handle requests.

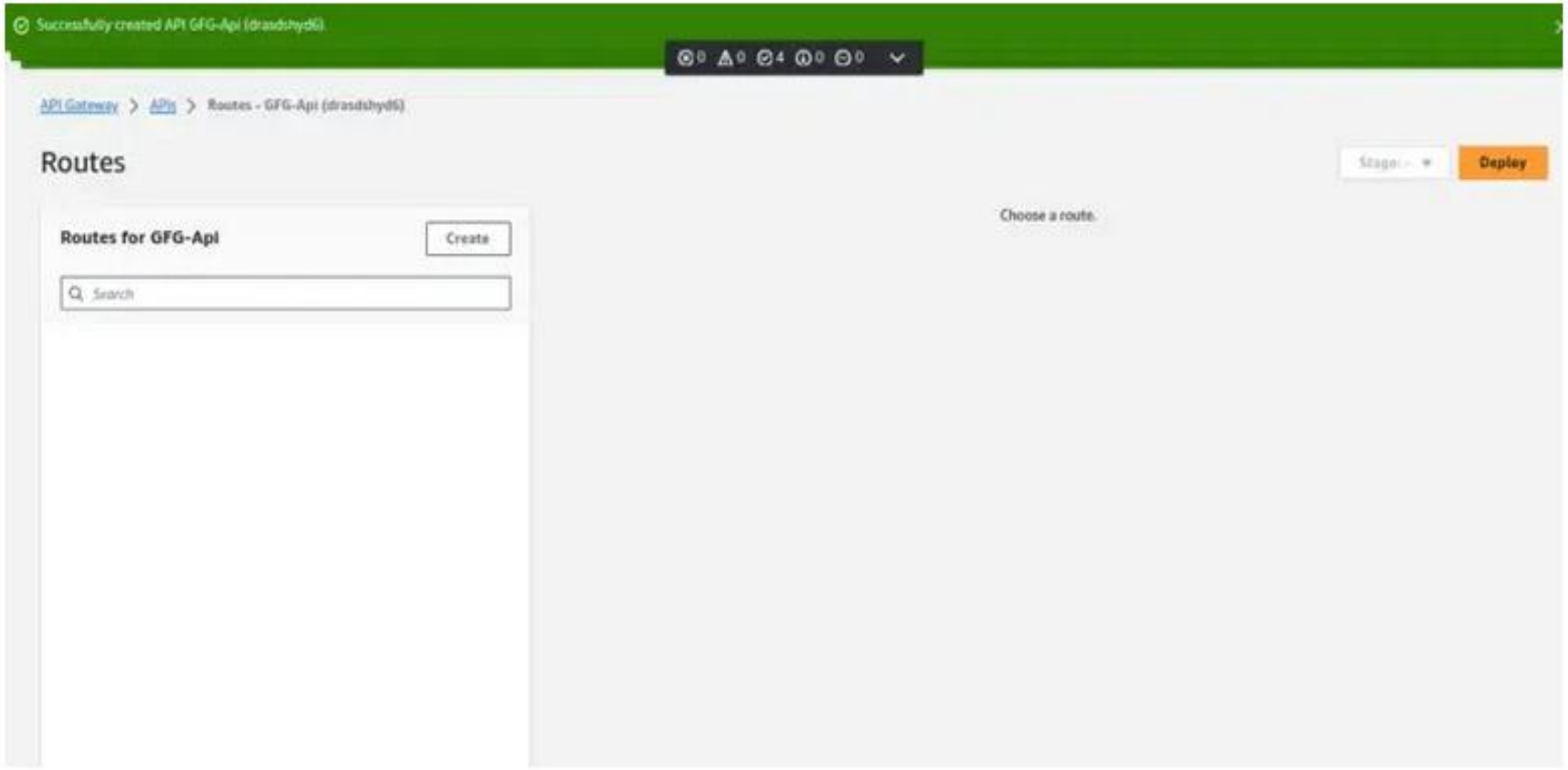
Step 4: Deploy the API

- Once the setup is complete, deploy your API Gateway



Step 5: Review and Create

- After reviewing all your settings and configurations, click Review and Create. Once the API is created, you can use it with the name you've assigned.



Amazon API Gateway Advantages

- Enables efficient API development by allowing multiple versions of the same API to run simultaneously, helping in quick testing and iteration.
- Provides very low latency for API requests and responses, improving overall performance.
- Offers easy performance monitoring through the API Gateway dashboard.
- Cost-efficient at scale, allowing users to reduce costs as API usage increases across regions and accounts.
- Provides flexible security controls using AWS IAM, Amazon Cognito, and custom authorizers.
- Reduces coding effort, making application development more efficient with fewer chances of errors.
- Hides and encapsulates the internal structure of backend applications from clients.
- Allows clients to interact through a single entry point instead of calling individual services directly.
- Simplifies the management and handling of multiple APIs in one centralized place.

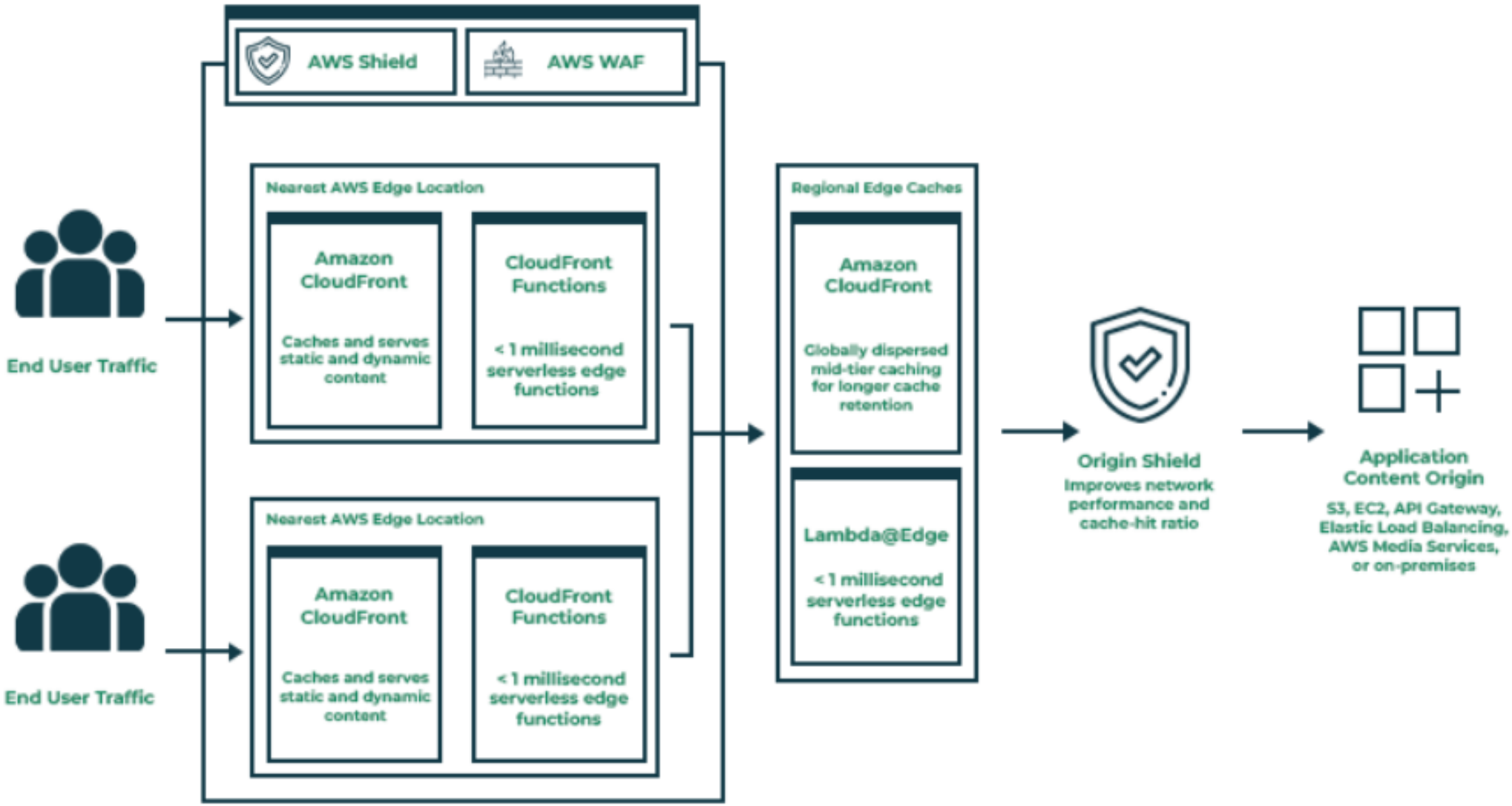
Amazon API Gateway Disadvantages

- Security implementation can be challenging, as applying security practices to every API call may become complex and make performance tracking difficult.
- API Gateway may not always be considered fully reliable due to dependency on a single gateway layer.
- Maintaining the API Gateway becomes difficult when an application grows into hundreds or thousands of microservices.
- Requires routing rules, which can create a single point of failure if not managed properly.
- Storing all API rules in one place increases the risk of complexity, especially as the number of services scales.

AWS CloudFront

AWS CloudFront Overview

- AWS CloudFront is a fast content delivery network (CDN) that speeds up the delivery of websites, videos, APIs, and applications.
- It works by storing copies of your content in multiple edge locations worldwide.
- When a user visits your site, CloudFront delivers content from the nearest edge location, reducing latency and improving load times.


AWS Shield
AWS WAF
Nearest AWS Edge Location
Amazon CloudFront
Caches and serves static and dynamic content
CloudFront Functions
< 1 millisecond serverless edge functions
Nearest AWS Edge Location
Amazon CloudFront
Caches and serves static and dynamic content
CloudFront Functions
< 1 millisecond serverless edge functions
End User Traffic
End User Traffic
Regional Edge Caches
Amazon CloudFront
Globally dispersed mid-tier caching for longer cache retention
Lambda@Edge
< 1 millisecond serverless edge functions
Origin Shield
Improves network performance and cache-hit ratio
Application Content Origin
S3, EC2, API Gateway, Elastic Load Balancing, AWS Media Services, or on-premises

What is AWS CloudFront?

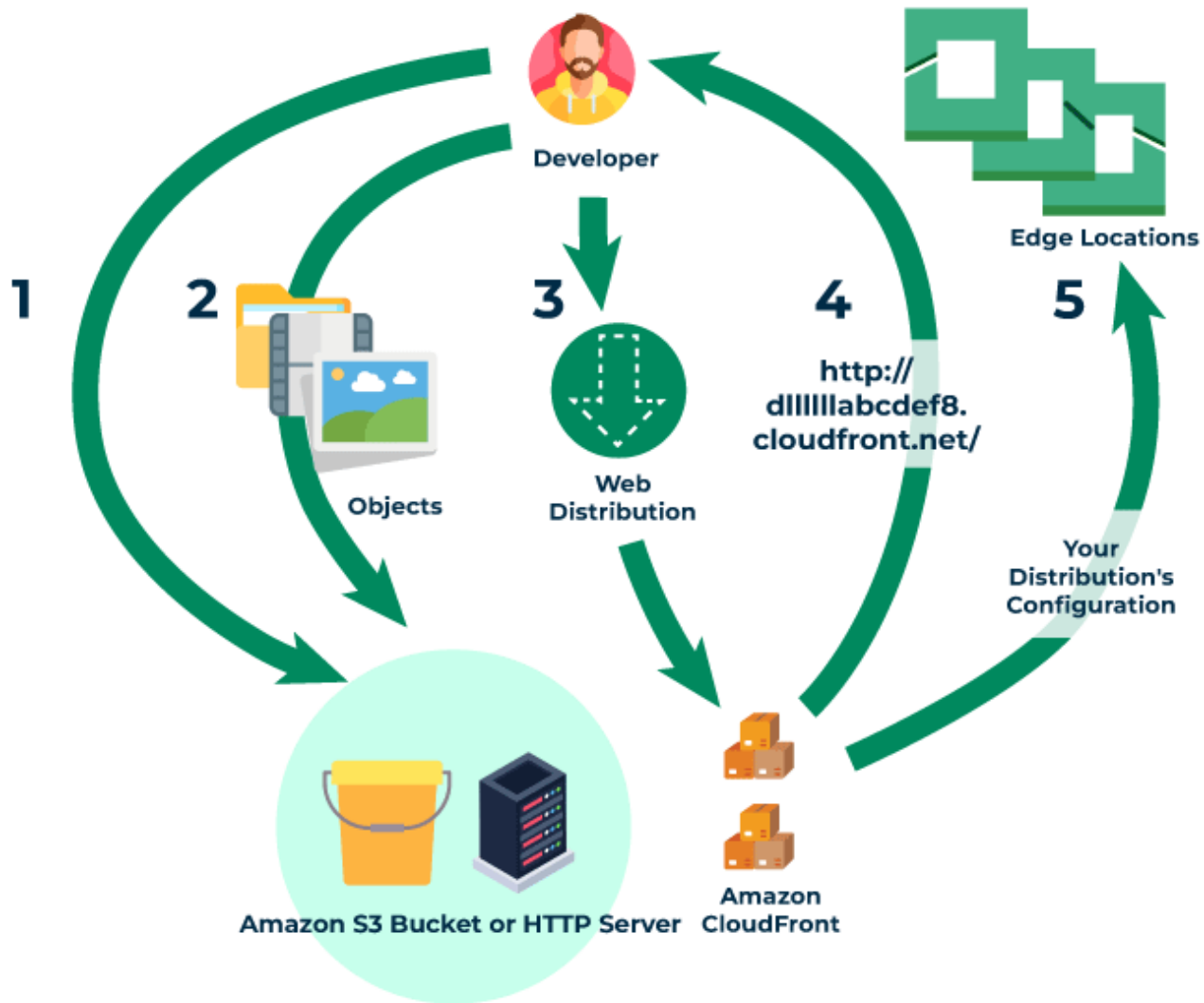
- AWS CloudFront is a global Content Delivery Network (CDN) service offered by Amazon Web Services.
- It helps deliver content—such as static files, dynamic data, videos, images, and APIs—to users quickly and efficiently across the world.
- CloudFront stores cached copies of your content in *edge locations*, which are servers placed globally for faster delivery.
- When a user makes a request, CloudFront delivers the content from the nearest edge location, reducing latency and improving performance.
- It integrates seamlessly with AWS services like Amazon S3, EC2, Lambda@Edge, and API Gateway.

Key Components of AWS CloudFront

- **Edge Locations:** Global data centers that cache content closer to users.
- **Origin Server:** The main location where your original content is stored (e.g., S3 bucket, EC2 instance, or on-premises server).
- **Distribution:** Configuration that defines how CloudFront delivers content.
  - Web Distribution (for websites, APIs, static/dynamic content)
  - RTMP Distribution (for streaming media; now deprecated)
- **Cache Behaviors:** Rules that define how content should be cached (TTL settings, cookies, query strings, etc.).
- **Signed URLs & Cookies:** Used to restrict access and protect private content.
- **Lambda@Edge:** Allows running custom logic at edge locations to modify requests/responses.

How AWS CloudFront Works

- **User Requests Content**
  - A user requests a webpage, image, video, or API response from a website or application.
- **DNS Routes the Request**
  - DNS identifies the nearest CloudFront edge location and routes the request there.
- **CloudFront Checks Cache**
  - If the content is already cached at the edge location → it is delivered immediately.
  - If not cached → CloudFront forwards the request to the origin server.
- **Origin Server Sends Content**
  - The origin (S3, EC2, or on-premises server) returns the requested content to CloudFront.
- **CloudFront Caches the Content**
  - CloudFront stores the new content in the edge location for future requests.
- **Content Delivered to the User**
  - The user receives the content quickly from the nearest edge location.
- **Faster Future Requests**
  - Next time, CloudFront delivers the cached version instantly without contacting the origin.
- **Cache Updates When Needed**
  - CloudFront checks the origin periodically. If the content has changed, it updates the cached version.


Developer
Edge Locations
1
2
3
4
5
Objects
Web Distribution
http://
dllllllabcdef8.
cloudfront.net/
Your Distribution's Configuration
Amazon S3 Bucket or HTTP Server
Amazon CloudFront

Key Features of AWS CloudFront

- **Faster Content Delivery Across the Globe**
  - CloudFront stores copies of content in many global edge locations.
  - Users receive content from the nearest location, ensuring faster loading times.
- **Seamless Integration with AWS Services**
  - Works smoothly with S3 (file storage), EC2 (web hosting), API Gateway (APIs), Route 53 (DNS), AWS WAF (security), and ELB (load balancing).
- **Built-in Security & Attack Protection**
  - Protects against DDoS attacks and malicious traffic through AWS Shield and AWS WAF.
  - Supports HTTPS and access control for secure content delivery.
- **Efficient Caching for Static & Dynamic Content**
  - Static files (images, CSS, JS) are cached and delivered instantly.
  - Dynamic content (APIs, personalized pages) is optimized for low-latency delivery.
- **Budget-Friendly & Scalable**
  - Pay only for what you use.
  - AWS Free Tier includes **1 TB** of data transfer per month.
- **Customizable with Lambda@Edge**
  - Modify request/response behavior at edge locations (e.g., header changes, redirects, A/B testing).
- **Low Latency & High Performance**
  - CloudFront finds the fastest route to deliver content, providing smoother user experiences.

AWS CloudFront & AWS WAF Integration

What is AWS WAF?

- A security service that protects web applications from threats like SQL injection and XSS.
- Filters harmful requests and allows you to set custom security rules.

How CloudFront Works with WAF

- You create **Web ACLs** (Access Control Lists) containing security rules.
- When a user requests content, CloudFront routes it to the nearest edge location.
- The Web ACL checks the request:
  - **If allowed:** The request goes to the origin.
  - **If blocked:** The request is stopped at the edge.

What is a Web ACL?

- A set of rules used to control traffic based on:
- IP addresses
- Request headers
- Query strings or request body
- Geographic locations

AWS CloudFront Use Cases

- **Delivering Static Web Content**
  - Static files (HTML, CSS, JS, images) are cached at edge locations for fast delivery.
- **Streaming Media**
  - Integrates with S3 to stream video and audio without buffering.
- **Serving Dynamic Content**
  - Lambda@Edge helps generate and deliver dynamic data efficiently.
- **Global Content Delivery**
  - Content is cached near users around the world, reducing latency and improving accessibility.

- **Key Benefits of AWS CloudFront**
- No mandatory upfront investment
- Reduced operational costs
- Highly scalable and resilient
- Easy global access
- Lower maintenance effort and reduced business risks

AWS CloudFront Pricing

- The table below provides a detailed knowledge of Cost of CloudFront:

Pricing Component	Description	Cost
Data Transfer Out	Data delivered from CloudFront to the internet.	Starting at $0.085 per GB for the first 10 TB/month in the U.S., Mexico, and Canada.
HTTP/HTTPS Requests	Number of requests processed by CloudFront.	$0.0075 per 10,000 HTTP requests; $0.0100 per 10,000 HTTPS requests in the U.S. region.
Invalidation Requests	Removing cached objects before expiration.	First 1,000 paths free each month; $0.005 per path thereafter.
Real-Time Log Requests	Detailed logging of CloudFront requests.	$0.01 per 1,000,000 log lines.
Origin Shield Requests	Additional caching layer to reduce origin load.	$0.0075 per 10,000 requests in the U.S. region.

Companies using AWS CloudFront

The following are list of companies using AWS CloudFront:

Company	Use Case
United States Department of Defense	Secure content distribution across its extensive network.
Walmart	Enhances performance and reliability of its online retail platform.
Amazon	Accelerates content delivery for its e-commerce operations.
Netflix	Streams high-quality video content globally with low latency.
Hulu	Delivers streaming media content efficiently to its users.
Disney	Supports Disney+ with scalable and secure content delivery.
BMW	Enhances vehicle connectivity and digital services via CloudFront.
Unilever	Powers digital marketing and global campaign deployment.
McDonald's	Optimizes customer experience with machine learning and analytics.
Capital One	Supports banking services with high-security cloud solutions.

Amazon Virtual Private Cloud

What is a VPC (Virtual Private Cloud)?

- A VPC is an on-demand, configurable pool of cloud resources within a public cloud environment.
- It provides a **logically isolated** and **secure** section of the cloud, similar to a private data center.
- Organizations can deploy applications and store data in a private virtual network that looks and works like an on-premises setup but benefits from the cloud's scalability and flexibility.
- A VPC ensures your resources stay separate from other cloud tenants, giving you full control over networking, security, and access.

Benefits of Using a VPC

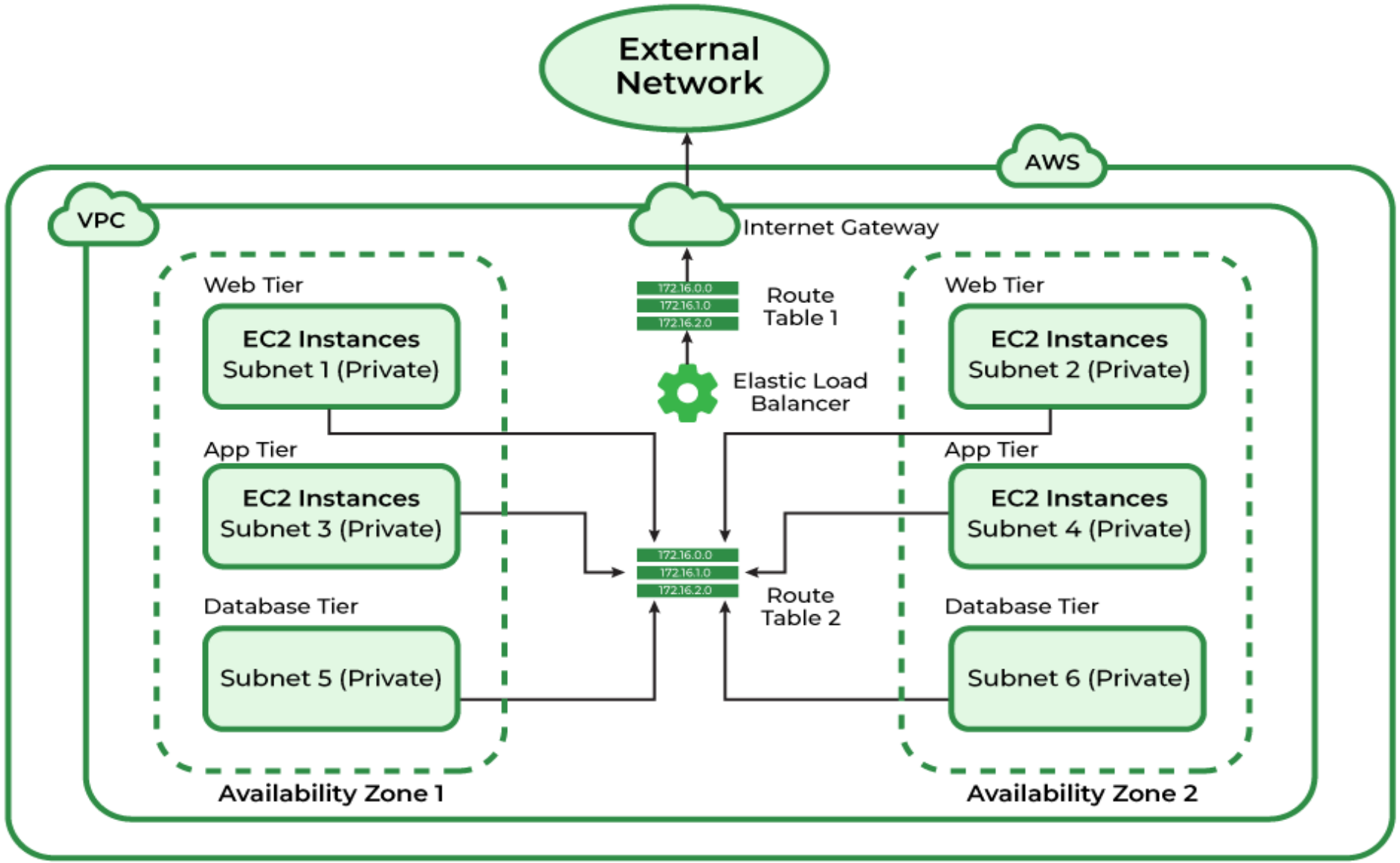
- **Security**
  - VPC ensures strong logical isolation of workloads.
  - Provides strict access controls to keep resources safe from unauthorized access.
  - Keeps your environment isolated from other customers.
- **Agility & Scalability**
  - Allows organizations to quickly deploy, modify, or scale resources.
  - No need to purchase or install physical hardware.
  - Network size and configuration can be changed anytime.
- **High Availability**
  - Uses the cloud provider’s fault-tolerant infrastructure.
  - Supports deployment across multiple Availability Zones to ensure redundancy.
  - Improves uptime and reliability of applications.
- **Cost Efficiency**
  - Offers the features of a private data center without high upfront investment.
  - Reduces hardware, maintenance, and labor costs.
  - Pay-as-you-go pricing helps optimize expenses.

Example

- **Office Building** → VPC (private and secure space)
- **Departments (Editorial, HR, Development)** → Subnets (public or private)
- **Security Guards** → Security Groups & Firewalls (control who enters or leaves)
- **Internet Connection** → Internet Gateway (allows public access when required)
- **Private Tunnels to Partners** → VPN / Direct Connect (secure external connections)

Amazon VPC Architecture Overview

- A VPC is built using multiple components like **Gateways**, **Subnets**, **Load Balancers**, and more.
- These resources combine to create a controlled and isolated virtual environment.
- Networking is managed using **route tables** that connect different subnets.
- Multiple layers of security—such as NACLs and Security Groups—protect the network.
- Public and private subnets allow separation of externally accessible and internal resources.


External Network
AWS
VPC
Internet Gateway
Web Tier
EC2 Instances
Subnet 1 (Private)
172.16.0.0
172.16.1.0
172.16.2.0
Route Table 1
Elastic Load Balancer
Web Tier
EC2 Instances
Subnet 2 (Private)
App Tier
EC2 Instances
Subnet 3 (Private)
App Tier
EC2 Instances
Subnet 4 (Private)
172.16.0.0
172.16.1.0
172.16.2.0
Route Table 2
Database Tier
Subnet 5 (Private)
Database Tier
Subnet 6 (Private)
Availability Zone 1
Availability Zone 2

Core Components of Amazon VPC

- The following are the key components of Amazon VPC:

1. VPC (Virtual Private Cloud)

- A logically isolated section of the AWS cloud where resources can be launched securely.
- Defined using an IPv4 CIDR block (e.g., **10.0.0.0/16**) providing up to **65,536 IP addresses**.
- Functions like an on-premises data center but with AWS scalability, reliability, and elasticity.

2. Subnets

- A subdivision of a VPC's IP range and exists in **one Availability Zone**.
- **Purpose:**
  - *Segmentation*: Organizes and isolates resources.
  - *High Availability*: Distributing subnets across AZs ensures fault tolerance.
- **Types:**
  - **Public Subnets** – route traffic through an Internet Gateway (IGW).
  - **Private Subnets** – isolated from direct internet access; use NAT Gateway for outbound traffic.

3. Route Tables

- Each VPC has an implicit virtual router that uses **Route Tables** to direct traffic.
- Every subnet must be associated with one route table.
- A route includes:
  - **Destination CIDR** (traffic target)
  - **Target** (IGW, NAT Gateway, instance, etc.)
- Determines whether traffic stays in the VPC or goes outside.

4. Network ACLs (NACLs)

- Stateless firewalls operating at the **subnet level**.
- Default NACL provided by AWS; modifiable but not deletable.
- Require separate rules for inbound and outbound traffic.
- Ideal for broad subnet-level restrictions like blocking specific IP ranges.

5. Internet Gateway (IGW)

- A horizontally scaled, redundant component enabling VPC resources to access the internet.
- Attached to a VPC to allow bidirectional communication.
- A public subnet must have a route like **0.0.0.0/0 → IGW** for internet access.
- Only **one IGW** can be attached to a VPC at a time.

6. Network Address Translation (NAT)

- **NAT Gateways** allow instances in private subnets to access the internet **outbound only**.
- Prevents unsolicited inbound traffic.
- Commonly placed in a **public subnet**.
- Ensures private instances can download updates, access APIs, etc., without exposing them.

7. Security Groups

- Stateful firewalls applied at the **instance level**.
- Allow inbound/outbound rules based on IP, port, and protocol.
- Return traffic for an allowed outbound request is automatically permitted.
- Provide granular control for EC2, RDS, Lambda functions, and other VPC resources.

8. CIDR (Classless Inter-Domain Routing)

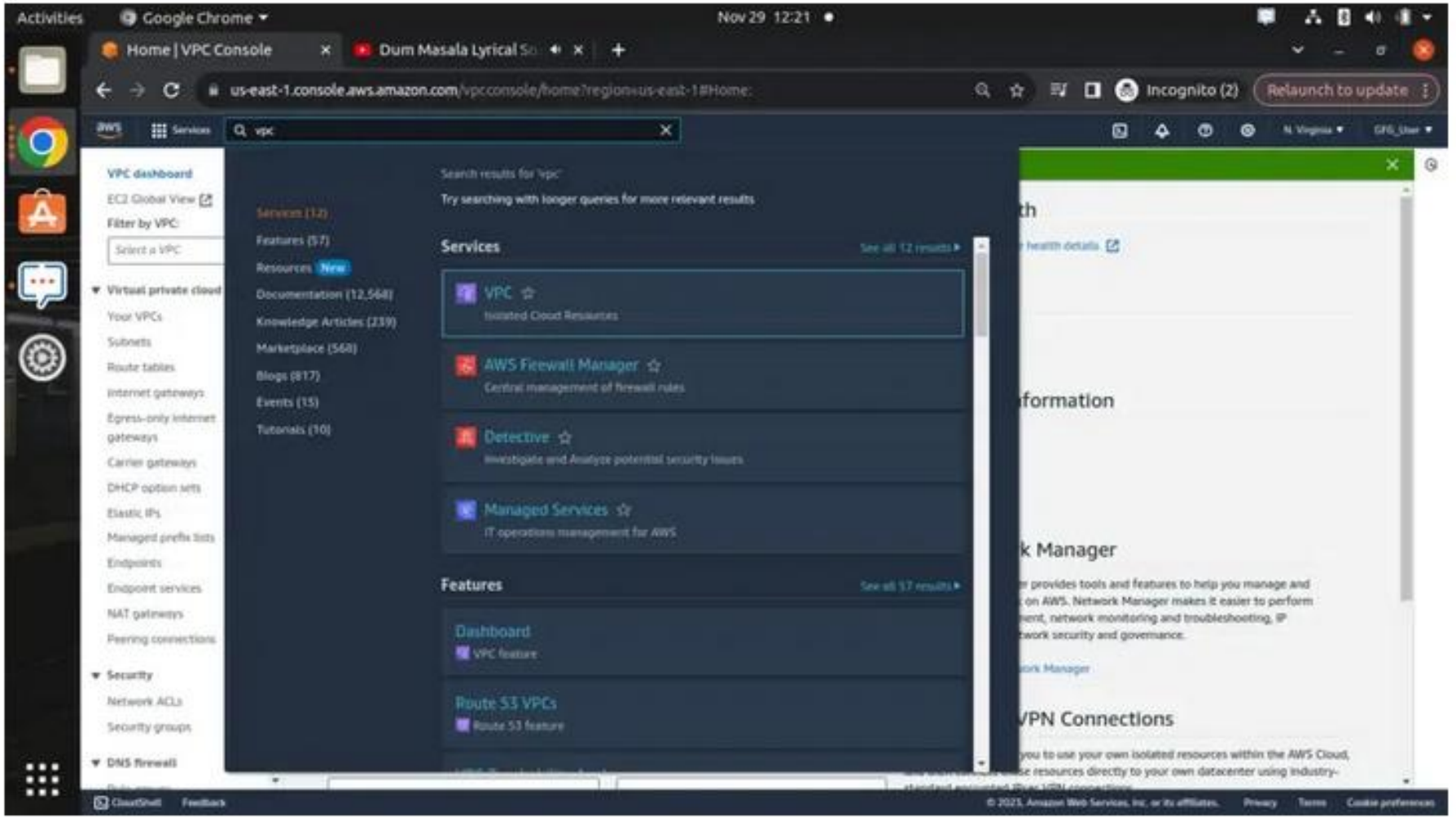
- Defines the IP addressing structure of a VPC or subnet (e.g., **10.0.0.0/16**).
- Supports flexible allocation and subdivision of private IP ranges per **RFC 1918**, including:
  - **10.0.0.0/8**
  - **172.16.0.0/12**
  - **192.168.0.0/16**

Use Cases of Amazon VPC

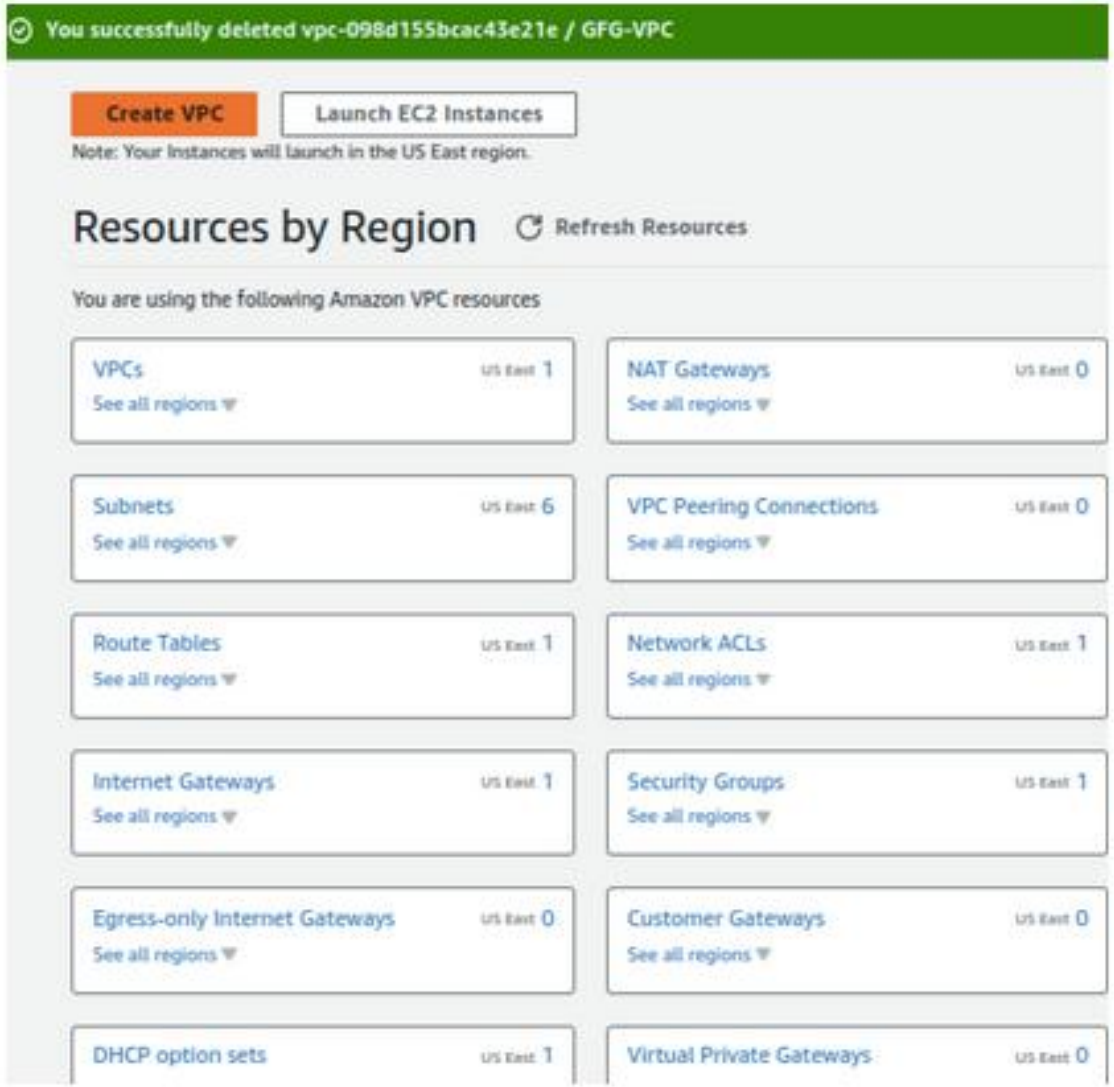
- Hosting public applications (web servers, ALBs) in public subnets.
- Deploying multi-tier architectures (app servers + databases) in private subnets.
- Secure inter-VPC connectivity via peering or AWS Transit Gateway.
- Enforcing security and compliance through isolation, monitoring, and access control (e.g., VPC Flow Logs, Security Groups, NACLs).

Amazon VPC Working – Steps

Step 1: Log in to the AWS Management Console and navigate to the **VPC Dashboard**.



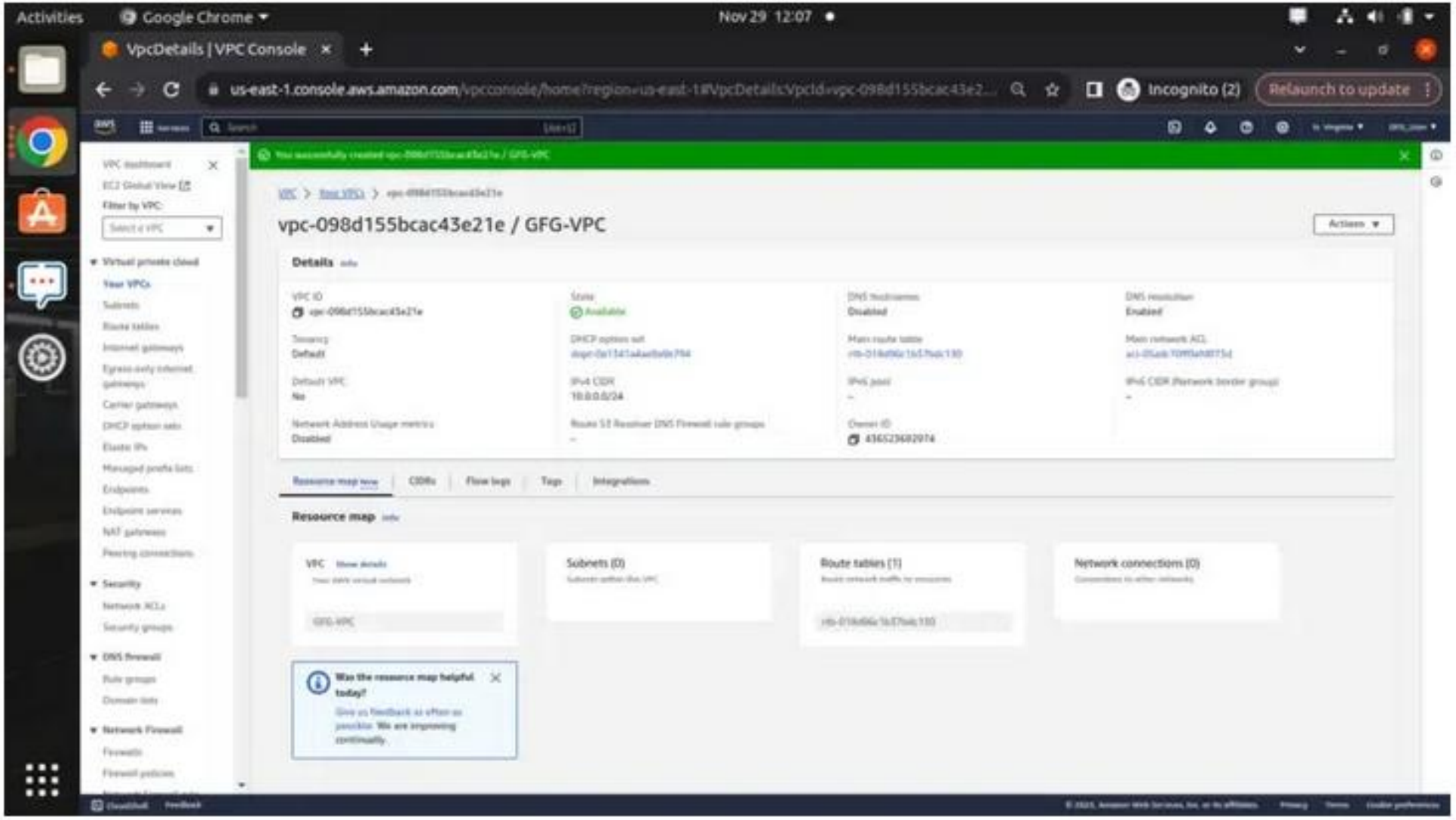
Step 2: After navigating to the AWS VPC now click on create VPC.



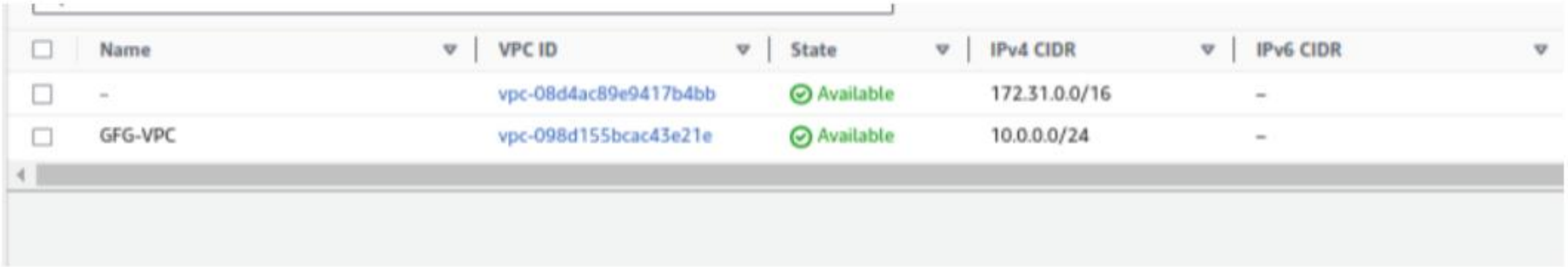
Step 3: Configure all the details required to create as shown in the image below. Some of the most required settings to configure VPC are as follows

- Name of the Network.
- IPv4 CIDR.
- And tags of VPC after that click on create VPC.

Step 4: Virtual Private Cloud created successfully with the required settings



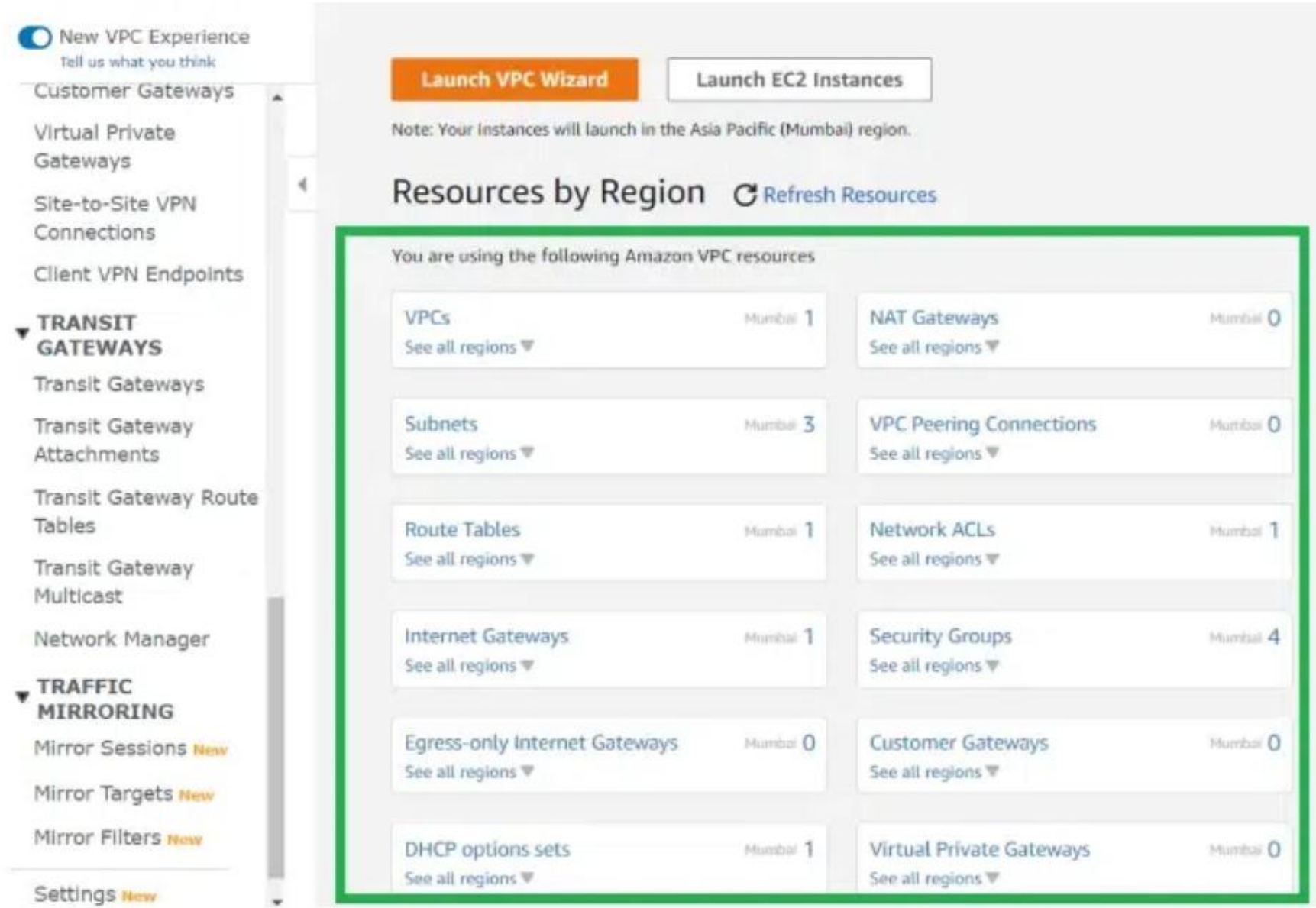
Step 5: Check the VPC dashboard whether the VPC created is available to use as shown in the image below GFG-VPC.



Name	VPC ID	State	IPv4 CIDR	IPv6 CIDR
-	vpc-08d4ac89e9417b4bb	Available	172.31.0.0/16	-
GFG-VPC	vpc-098d155bcac43e21e	Available	10.0.0.0/24	-

AWS VPC Console

- To create and manage a Virtual Private Cloud (VPC) in AWS, follow these steps:
- **Log in** to your AWS account.
- Once inside the **AWS Management Console**, click on the **“Services”** menu at the top.
- From the list of categories, navigate to **“Networking & Content Delivery”**.
- Select **“VPC”** from the options provided.
- After selecting VPC, you will be redirected to the VPC dashboard. On the left-hand side, the **navigation pane** displays various options and services related to VPC management.


New VPC Experience
Tell us what you think
Customer Gateways
Virtual Private Gateways
Site-to-Site VPN Connections
Client VPN Endpoints
TRANSIT GATEWAYS
Transit Gateways
Transit Gateway Attachments
Transit Gateway Route Tables
Transit Gateway Multicast
Network Manager
TRAFFIC MIRRORING
Mirror Sessions New
Mirror Targets New
Mirror Filters New
Settings New
Launch VPC Wizard
Launch EC2 Instances
Note: Your instances will launch in the Asia Pacific (Mumbai) region.
Resources by Region
Refresh Resources
You are using the following Amazon VPC resources
VPCs Mumbai 1 See all regions
NAT Gateways Mumbai 0 See all regions
Subnets Mumbai 3 See all regions
VPC Peering Connections Mumbai 0 See all regions
Route Tables Mumbai 1 See all regions
Network ACLs Mumbai 1 See all regions
Internet Gateways Mumbai 1 See all regions
Security Groups Mumbai 4 See all regions
Egress-only Internet Gateways Mumbai 0 See all regions
Customer Gateways Mumbai 0 See all regions
DHCP options sets Mumbai 1 See all regions
Virtual Private Gateways Mumbai 0 See all regions

Demonstrate the working principles of Amazon VPC (Virtual Private Cloud) and explain the VPC subnets and Route tables.

INTRODUCTION

- Amazon VPC creates a **logically isolated network** in the AWS Cloud.
- Provides **full control** over virtual networking.
- **Key features are:**
- ***IP Address Ranges***: Define IPv4 CIDR blocks for the VPC.
- ***Subnet Creation***: Logical divisions within the VPC.
- ***Route Tables and NAT configuration***: Manage traffic flow within and outside the VPC.
- ***Security***: Utilize security groups and Network Access Control Lists (NACLs) for instance-level and subnet-level protection.

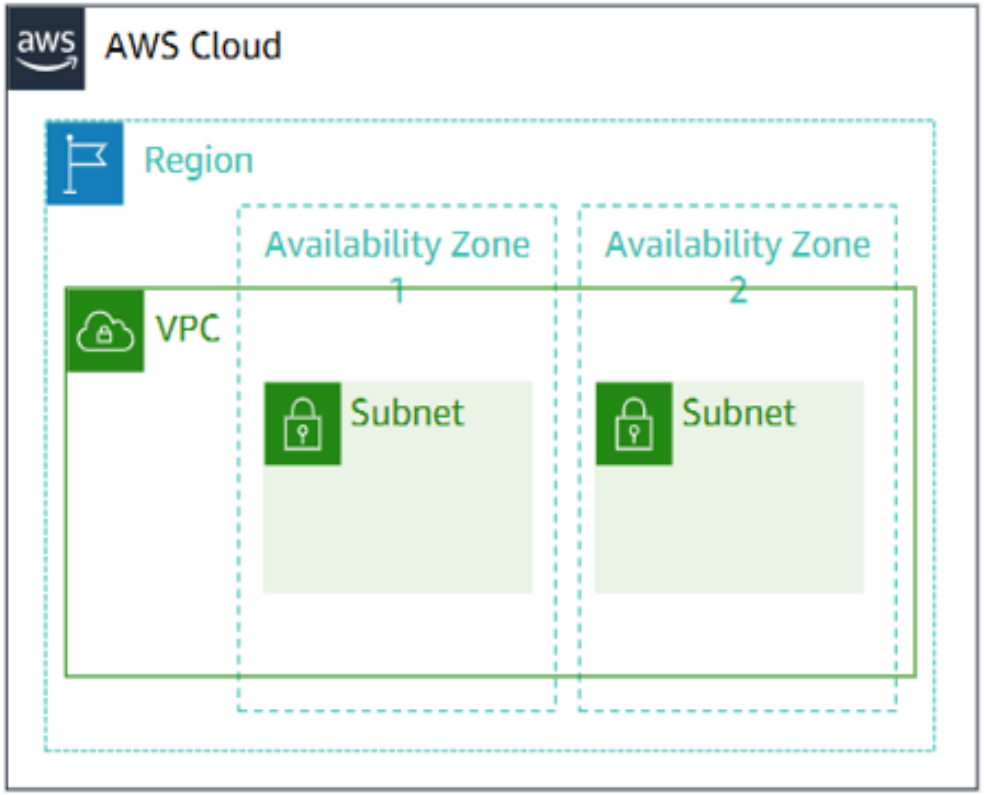
WORKING PRINCIPLES

VPC:

- Logically isolated from other VPCs.
- Dedicated to our AWS account.
- Belongs to a single AWS Region and can span multiple AZ.

Subnets:

- Range of IP addresses that divide a VPC.
- Belongs to a single AZ.
- Classified as Public or Private.



VPC SUBNET

- A **subnet** is a logical subdivision of an IP network.
- Dividing a network into smaller networks is called **subnetting**.
- **Subnetting** reduces network load and improves security.
- An **IP address** has two parts:
- **Network address** → common for all IPs in the group.

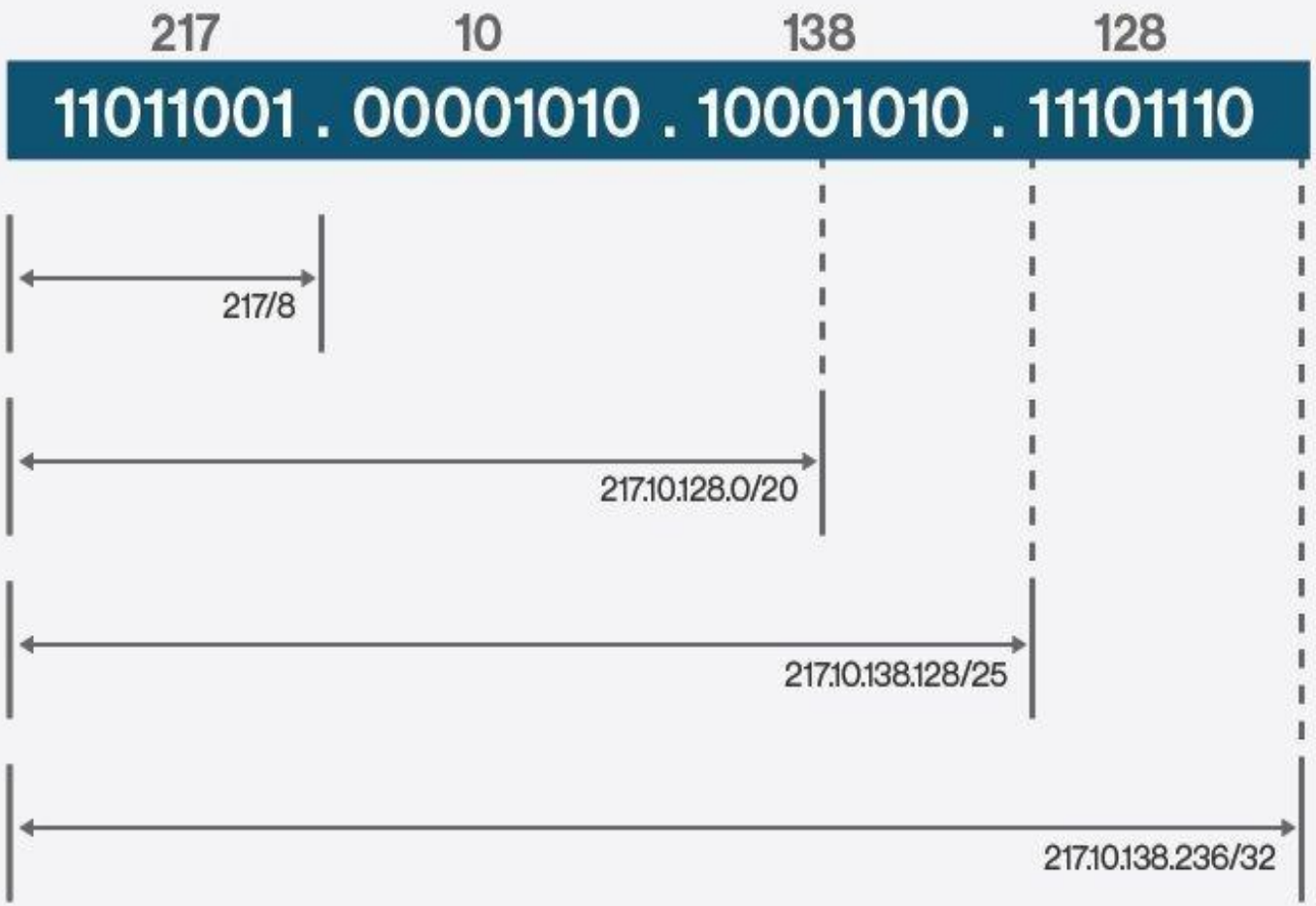
- **Host address** → unique for each device in the group.
- A **network address** = group address.
- A **host address** = node address.
- An **IP group** is called an **IP subnet**.
- **IP address** and **Subnet Mask** are both **32 bits long**.
- They are divided into **4 parts** (e.g., 125.3.2.1).
- A **Subnet Mask** assigns one bit for each IP bit (e.g., 255.255.255.0).
- **Subnet Mask bits** can be:
  - **ON (1)** → Network bit in the IP address.
  - **OFF (0)** → Host bit in the IP address.

Examples:

- IP: **10.10.10.10** → Subnet Mask: **255.0.0.0**
- IP: **171.154.12.5** → Subnet Mask: **255.255.0.0**
- IP: **192.163.1.1** → Subnet Mask: **255.255.255.0**
- **If the subnet mask is: 255.255.0.0 has 8-bits for subnet and 16-bits for Host.**
- **8-bits accommodate 2^8 = 256 subnets and 2^{16} = 64000 Hosts.**

CIDR - Classless Inter-Domain Routing

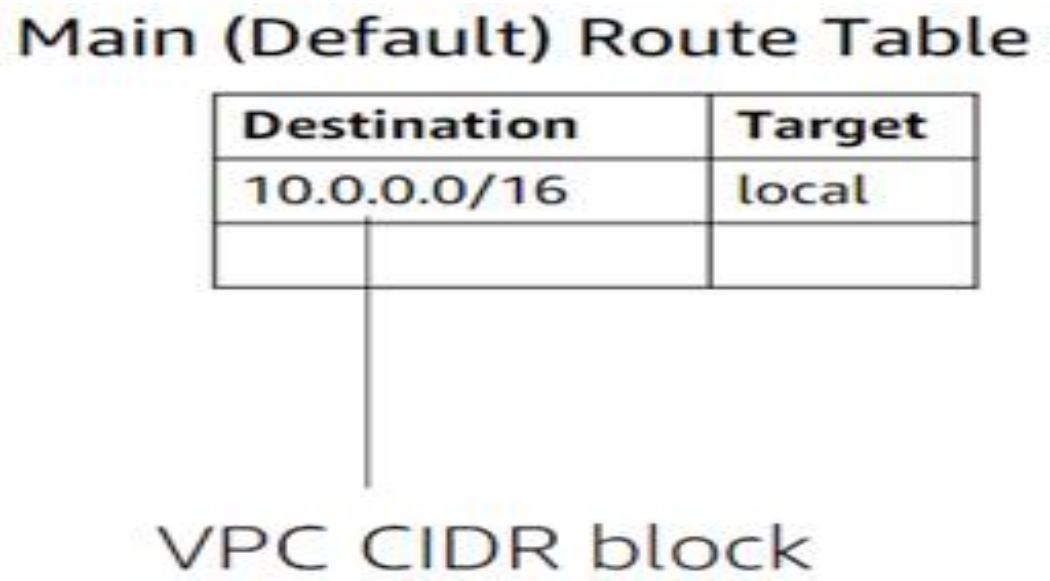
- Uses **VLSM (Variable-Length Subnet Mask)** to adjust Network/Host bit ratio.
- Allocates IP addresses and routes IP packets.
- Removes fixed IPv4 class system (Class A, B, C).
- Improves **IP address distribution efficiency**.
- **CIDR address format** has two parts:
  - **Prefix** → represents the Network address (like an IP).
  - **Suffix** → total number of bits in the address.
- **Ex: 192.168.129.23/17**
- **IPv4 addresses support a maximum of 32-bits.**
- **IPv6 addresses support a maximum of 128-bits.**



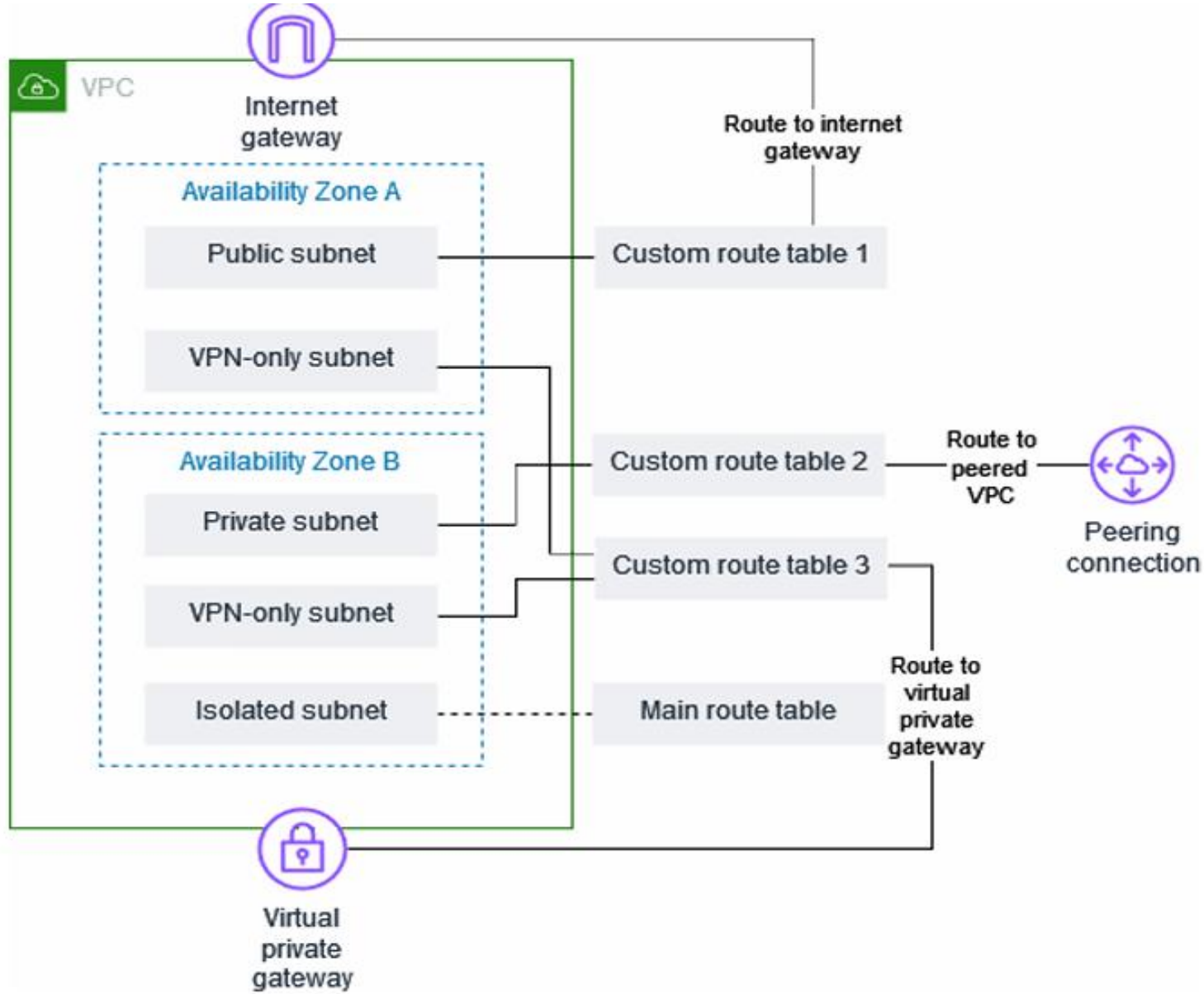
Source: Technology Training Ltd.

ROUTE TABLES

- A **route table** has rules to control subnet traffic.
- Each **route** has a **destination** and a **target**.
- By default, it includes a **local route** for communication inside the VPC.
- Each **subnet** must be linked to **one route table**.



- **Destination** = target CIDR block.
- **Target** = resource handling the traffic.
- **Public subnets** use an **Internet Gateway**.
- **Private subnets** use a **NAT Gateway** for internet access.


VPC
Internet gateway
Route to internet gateway
Availability Zone A
Public subnet
Custom route table 1
VPN-only subnet
Availability Zone B
Private subnet
Custom route table 2
Route to peered VPC
Peering connection
Custom route table 3
VPN-only subnet
Route to virtual private gateway
Isolated subnet
Main route table
Virtual private gateway

Key Concepts:

- **Main Route Table**: Default route table in a VPC; used by subnets not linked to another table.
- **Custom Route Table**: A route table created by the user.
- **Destination**: IP range where traffic should go (e.g., 172.16.0.0/12).
- **Target**: Gateway or resource handling the traffic (e.g., Internet Gateway).
- **Local Route**: Default route inside the VPC (exists for both IPv4 and IPv6).
- **Route Table Association**: Links a route table with a subnet, internet gateway, or virtual private gateway (VPG).
- **Subnet Route Table**: Route table connected to a subnet.
- **Propagation**: Automatically adds/removes VPN routes when a VPG is attached.
- **Gateway Route Table**: Route table linked to an Internet Gateway or VPG.
- **Edge Association**: Used to route inbound VPC traffic to an appliance.
- **Transit Gateway Route Table**: Route table linked to a transit gateway.
- **Local Gateway Route Table**: Route table linked to an Outposts local gateway.

Example VPC with Route Tables

- The VPC has **5 subnets**.
- It includes **1 main route table** and **3 custom route tables**.
- All **4 route tables** have **local routes**.
- **Custom route table-1** → route to **Internet Gateway**, linked to **public subnet in AZ-A**.
- **Custom route table-2** → route to **peered VPC**, linked to **private subnet in AZ-B**.
- **Custom route table-3** → route to **Virtual Private Gateway**, linked to **VPN-only subnets** in both AZs.

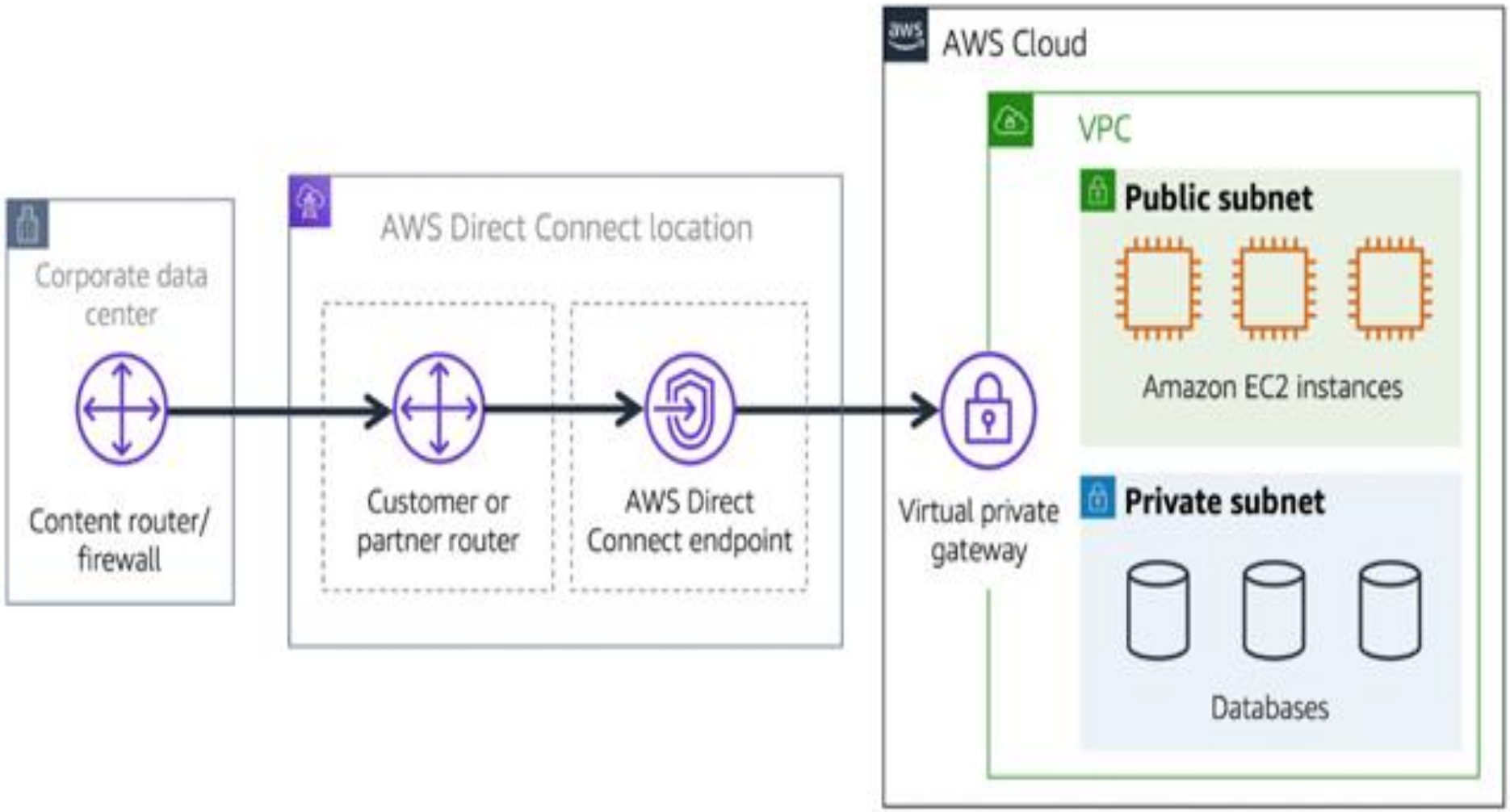
AWS Direct Connect

AWS Direct Connect

- **AWS Direct Connect** allows you to establish a **dedicated, private connection** between your on-premises data center and your AWS VPC.
- This private link **does not use the public internet**, offering greater security, consistent performance, and lower latency.
- It is ideal when you need **high bandwidth**, **stable connectivity**, and **secure communication** for enterprise workloads.

Analogy

- Imagine an **apartment building** directly connected to a **coffee shop** with a **private hallway**.
- Only the apartment residents can use this hallway—no one else.
- This private hallway represents **AWS Direct Connect**.
- Residents (your data center users) reach the coffee shop (AWS resources) **without using public roads** (public internet).
- Result: **Faster, safer, and exclusive access**.


AWS Cloud
VPC
Public subnet
Amazon EC2 instances
Private subnet
Databases
Virtual private gateway
AWS Direct Connect location
Customer or partner router
AWS Direct Connect endpoint
Corporate data center
Content router/ firewall

- The **private connection** provided by AWS Direct Connect helps **reduce network costs** by avoiding data transfer over the public internet.
- It allows you to **increase available bandwidth**, supporting large-scale data transfers and high-performance workloads.
- Direct Connect helps organizations meet **strict regulatory and compliance requirements** by providing a secure, dedicated, non-internet-based connection.
- It also helps avoid **potential bandwidth bottlenecks** or fluctuations that commonly occur on public internet connections.
- A **single VPC can have multiple types of gateways**, such as:
  - Internet Gateway
  - NAT Gateway
  - Virtual Private Gateway (AWS VPN)
  - Direct Connect Gateway
- These gateways support different types of resources and use cases, even if they all exist **within the same VPC** but reside in different subnets.

AWS Networking and Content Delivery

Content Delivery Network (CDN)

- **AWS Networking and Content Delivery** is a collection of cloud services that help you **connect**, **secure**, and **speed up** your applications and data on AWS. These services ensure that your applications communicate efficiently, safely, and with minimal latency across the globe.

Benefits of CDN

The following are the key benefits of CDN:

1. Improved Website Performance

- CDNs reduce latency by caching content at edge locations close to users, resulting in faster load times and a smoother browsing experience. This is crucial for maintaining user engagement and satisfaction.

2. Enhanced Reliability

- CDNs distribute traffic across multiple servers, ensuring that even if one server goes down, others can handle the load. This redundancy enhances the availability and reliability of websites and online services.|

3. Scalability

- CDNs can handle sudden spikes in traffic by distributing the load across their network. This scalability is essential for websites that experience variable traffic patterns, such as during product launches or viral content.

4. Security

- CDNs offer protection against DDoS attacks, provide secure data transmission through SSL/TLS, and can include additional security features like web application firewalls (WAF) to safeguard content and user data.

5. Cost Efficiency

- By offloading traffic from the origin server and reducing bandwidth consumption, CDNs can help lower infrastructure and operational costs. They also minimize the need for additional server capacity to handle peak loads.

CloudFront v/s Other CDN

The following table shows the key differences between CloudFront and other CDN:

Aspect	CloudFront	CDN
Service Provider	Amazon Web Services (AWS)	Various providers like Akamai, Cloudflare
Integration	Tightly integrated with AWS services	Compatible with various hosting environments
Customization	Offers extensive customization options	Provides basic to advanced customization

- AWS Networking and Content Delivery services help with:
- **Building private networks** inside AWS
- **Connecting on-premises data centers** to AWS
- **Routing traffic** between users and applications
- **Delivering content globally at high speed**
- **Protecting applications** from attacks and unauthorized access

Key Purposes

- **1. Connectivity**
- Connect your AWS resources securely and efficiently using:
- **VPC (Virtual Private Cloud)**
- **VPN**
- **AWS Direct Connect**

2. Traffic Management

- Control how users reach your applications using:
- **Elastic Load Balancing (ELB)**
- **Amazon Route 53 (DNS)**

3. Content Delivery

- Deliver content like images, videos, and APIs faster through:
- **Amazon CloudFront (CDN)**

4. Security

- Protect network traffic and applications using:
- **AWS WAF (Web Application Firewall)**
- **AWS Shield (DDoS protection)**
- **Security Groups & NACLs**